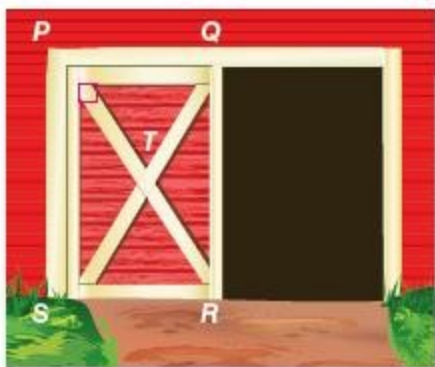


8-4 Rectangles

FARMING An X-brace on a rectangular barn door is both decorative and functional. It helps to prevent the door from warping over time.

If $ST = 3\frac{13}{16}$ feet, $PS = 7$ feet, and $m\angle PTQ = 67^\circ$, find each measure.



1. QR

SOLUTION:

The opposite sides of a rectangle are parallel and congruent. Therefore, $QR = PS = 7$ ft.

2. SQ

SOLUTION:

The diagonals of a rectangle bisect each other. So,

$$\begin{aligned} SQ &= 2(ST) \\ &= 2\left(3\frac{13}{16}\right) \\ &= \frac{61}{8} \\ &= 7\frac{5}{8} \end{aligned}$$

3. $m\angle TQR$

SOLUTION:

The diagonals of a rectangle are congruent and bisect each other. So, $\triangle TQR$ is an isosceles triangle.

Then, $\angle TQR \cong \angle TRQ$.

By the Exterior Angle Theorem,

$$m\angle TQR + m\angle TRQ = m\angle PTQ = 67^\circ.$$

$$\text{Therefore, } m\angle TQR = \frac{1}{2}(m\angle PTQ) = \frac{1}{2}(67^\circ) = 33.5^\circ.$$

4. $m\angle TSR$

SOLUTION:

The diagonals of a rectangle are congruent and bisect each other. So, $\triangle TSR$ is an isosceles triangle.

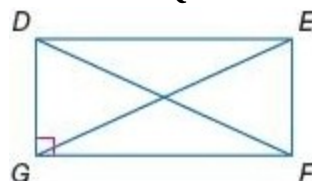
Then, $\angle TSR \cong \angle TRS$.

By the Vertical Angle Theorem,

$$m\angle STR = m\angle PTQ = 67^\circ.$$

$$\text{Therefore, } m\angle TSR = \frac{1}{2}(180^\circ - m\angle STR) = \frac{1}{2}(113^\circ) = 56.5^\circ.$$

ALGEBRA Quadrilateral $DEFG$ is a rectangle.



5. If $FD = 3x - 7$ and $EG = x + 5$, find EG .

SOLUTION:

The diagonals of a rectangle are congruent to each other. So, $FD = EG$.

$$3x - 7 = x + 5$$

$$2x = 12$$

$$x = 6$$

Use the value of x to find EG .

$$EG = 6 + 5 = 11$$

6. If $m\angle EFD = 2x - 3$ and $m\angle DFG = x + 12$, find $m\angle EFD$.

SOLUTION:

All four angles of a rectangle are right angles. So,

$$m\angle EFD + m\angle DFG = 90^\circ.$$

$$2x - 3 + x + 12 = 90$$

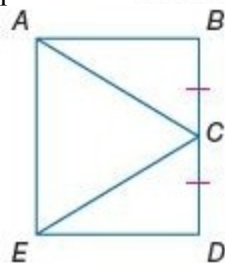
$$3x = 81$$

$$x = 27$$

$$\begin{aligned} m\angle EFD &= 2(27) - 3 \\ &= 51 \end{aligned}$$

8-4 Rectangles

7. **PROOF** If $ABDE$ is a rectangle and $\overline{BC} \cong \overline{DC}$, prove that $\overline{AC} \cong \overline{EC}$.

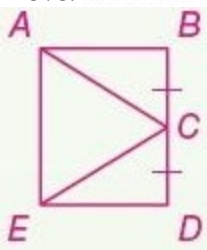


SOLUTION:

You need to walk through the proof step by step. Look over what you are given and what you need to prove. You are given $ABDE$ is a rectangle and $\overline{BC} \cong \overline{DC}$. You need to prove $\overline{AC} \cong \overline{EC}$. Use the properties that you have learned about rectangles to walk through the proof.

Given: $ABDE$ is a rectangle; $\overline{BC} \cong \overline{DC}$

Prove: $\overline{AC} \cong \overline{EC}$



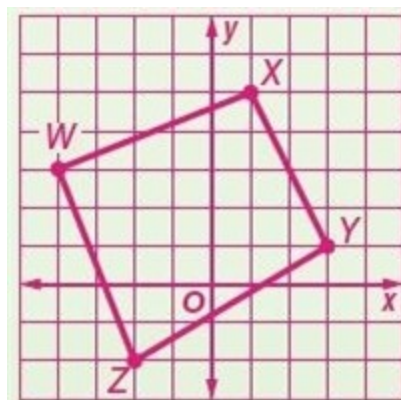
Statements(Reasons)

- $ABDE$ is a rectangle; $\overline{BC} \cong \overline{DC}$.
(Given)
- $ABDE$ is a parallelogram. (Def. of rectangle)
- $\overline{AB} \cong \overline{DE}$ (Opp. sides of a $\square$ are $\cong$.)
- $\angle B$ and $\angle D$ are right angles. (Def. of rectangle)
- $\angle B \cong \angle D$ (All rt $\angle$ s are $\cong$.)
- $\triangle ABC \cong \triangle EDC$ (SAS)
- $\overline{AC} \cong \overline{EC}$ (CPCTC)

COORDINATE GEOMETRY Graph each quadrilateral with the given vertices. Determine whether the figure is a rectangle. Justify your answer using the indicated formula.

8. $W(-4, 3)$, $X(1, 5)$, $Y(3, 1)$, $Z(-2, -2)$; Slope Formula

SOLUTION:



Use the slope formula to find the slope of the sides of the quadrilateral.

$$m_{WX} = \frac{5-3}{1-(-4)} = \frac{2}{5}$$

$$m_{XY} = \frac{1-5}{3-1} = -2$$

$$m_{YZ} = \frac{-2-1}{-2-3} = \frac{3}{5}$$

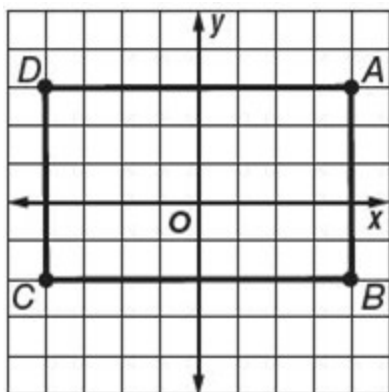
$$m_{WZ} = \frac{-2-3}{-2-(-4)} = -\frac{5}{2}$$

There are no parallel sides, so $WXYZ$ is not a parallelogram. Therefore, $WXYZ$ is not a rectangle.

8-4 Rectangles

9. $A(4, 3)$, $B(4, -2)$, $C(-4, -2)$, $D(-4, 3)$; Distance Formula

SOLUTION:



First, Use the Distance formula to find the lengths of the sides of the quadrilateral.

$$AB = \sqrt{(4 - 4)^2 + (-2 - 3)^2} = \sqrt{0 + (-5)^2} = \sqrt{25} = 5$$

$$BC = \sqrt{(-4 - 4)^2 + (-2 - (-2))^2} = \sqrt{(-8)^2 + 0} = \sqrt{64} = 8$$

$$CD = \sqrt{(-4 - (-4))^2 + (3 - (-2))^2} = \sqrt{0 + 5^2} = \sqrt{25} = 5$$

$$AD = \sqrt{(-4 - 4)^2 + (3 - 3)^2} = \sqrt{(-8)^2 + 0} = \sqrt{64} = 8$$

$AB = 5 = CD$, $BC = 8 = AD$, so $ABCD$ is a parallelogram.

A parallelogram is a rectangle if the diagonals are congruent. Use the Distance formula to find the lengths of the diagonals.

$$AC = \sqrt{(-4 - 4)^2 + (-2 - 3)^2} = \sqrt{(-8)^2 + (-5)^2} = \sqrt{64 + 25} = \sqrt{89}$$

$$BD = \sqrt{(4 - (-4))^2 + (3 - (-2))^2} = \sqrt{(8)^2 + (5)^2} = \sqrt{64 + 25} = \sqrt{89}$$

So the diagonals are congruent. Thus, $ABCD$ is a rectangle.

FENCING X-braces are also used to provide support in rectangular fencing. If $AB = 6$ feet, $AD = 2$ feet, and $m\angle DAE = 65^\circ$, find each measure.



10. BC

SOLUTION:

The opposite sides of a rectangle are parallel and congruent. Therefore, $BC = AD = 2$ ft.

11. DB

SOLUTION:

All four angles of a rectangle are right angles. So, $\triangle ABC$ is a right triangle. By the Pythagorean

Theorem, $DB^2 = BC^2 + DC^2$.

$$BC = AD = 2$$

$$DC = AB = 6$$

$$DB^2 = (2)^2 + (6)^2 = 40$$

$$DB = \sqrt{40} \approx 6.3$$

12. $m\angle CEB$

SOLUTION:

The diagonals of a rectangle are congruent and bisect each other. So, $\triangle AED$ is an isosceles triangle.

Then, $\angle DAE \cong \angle ADE$.

$$m\angle AED = 180 - (m\angle DAE + m\angle ADE)$$

$$= 180 - (65 + 65)$$

$$= 50$$

By the Vertical Angle Theorem,

$$m\angle CEB = m\angle AED = 50.$$

13. $m\angle EDC$

SOLUTION:

The diagonals of a rectangle are congruent and bisect each other. So, $\triangle AED$ is an isosceles triangle.

Then, $m\angle ADE = m\angle DAE = 50$.

All four angles of a rectangle are right angles.

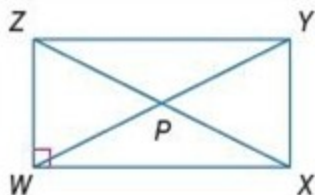
$$m\angle EDC = 90 - m\angle ADE$$

$$= 90 - 65$$

$$= 25$$

8-4 Rectangles

CCSS REGULARITY Quadrilateral $WXYZ$ is a rectangle.



14. If $ZY = 2x + 3$ and $WX = x + 4$, find WX .

SOLUTION:

The opposite sides of a rectangle are parallel and congruent. So, $ZY = WX$.

$$2x + 3 = x + 4$$

$$x = 1$$

Use the value of x to find WX .

$$WX = 1 + 4 = 5.$$

15. If $PY = 3x - 5$ and $WP = 2x + 11$, find ZP .

SOLUTION:

The diagonals of a rectangle bisect each other. So, $PY = WP$.

$$3x - 5 = 2x + 11$$

$$x = 16$$

The diagonals of a rectangle are congruent and bisect each other. So, $\triangle ZPW$ is an isosceles triangle.

$$ZP = WP$$

$$= 2(16) + 11$$

$$= 43$$

16. If $m\angle ZYW = 2x - 7$ and $m\angle WYX = 2x + 5$, find $m\angle ZYW$.

SOLUTION:

All four angles of a rectangle are right angles. So,

$$m\angle ZYW + m\angle WYX = 90.$$

$$2x - 7 + 2x + 5 = 90$$

$$4x = 92$$

$$x = 23$$

$$m\angle ZYW = 2(23) - 7$$

$$= 39$$

17. If $ZP = 4x - 9$ and $PY = 2x + 5$, find ZX .

SOLUTION:

The diagonals of a rectangle are congruent and bisect each other. So, $ZP = PY$.

$$4x - 9 = 2x + 5$$

$$2x = 14$$

$$x = 7$$

$$\text{Then } ZP = 4(7) - 9 = 19.$$

$$\text{Therefore, } ZX = 2(ZP) = 38.$$

18. If $m\angle XZY = 3x + 6$ and $m\angle XZW = 5x - 12$, find $m\angle YXZ$.

SOLUTION:

All four angles of a rectangle are right angles. So,

$$m\angle YZX + m\angle XZW = 90.$$

$$3x + 6 + 5x - 12 = 90$$

$$8x = 96$$

$$x = 12$$

$$m\angle YXZ = m\angle XZW = 5(12) - 12 = 48$$

19. If $m\angle ZXW = x - 11$ and $m\angle WZX = x - 9$, find $m\angle ZXY$.

SOLUTION:

All four angles of a rectangle are right angles. So,

$\triangle ZWX$ is a right triangle. Then,

$$m\angle ZXW + m\angle WZX = 90.$$

$$x - 11 + x - 9 = 90$$

$$2x = 110$$

$$x = 55$$

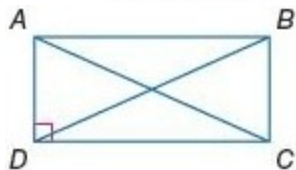
Therefore, $m\angle WZX = 55 - 9$ or 46. Since $\overline{ZX}$ is a transversal of parallel sides $\overline{ZY}$ and $\overline{WX}$, alternate interior angles $\angle WZX$ and $\angle ZXY$ are congruent. So, $m\angle ZXY = m\angle WZX = 46$.

8-4 Rectangles

PROOF Write a two-column proof.

20. **Given:** $ABCD$ is a rectangle.

Prove: $\triangle ADC \cong \triangle BCD$



SOLUTION:

You need to walk through the proof step by step. Look over what you are given and what you need to prove. Here, you are given $ABCD$ is a rectangle. You need to prove $\triangle ADC \cong \triangle BCD$. Use the properties that you have learned about rectangles to walk through the proof.

Given: $ABCD$ is a rectangle.

Prove: $\triangle ADC \cong \triangle BCD$

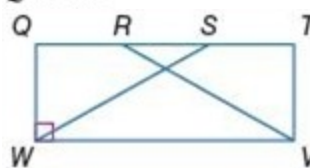
Proof:

Statements (Reasons)

1. $ABCD$ is a rectangle. (Given)
2. $ABCD$ is a parallelogram. (Def. of rectangle)
3. $\overline{AD} \cong \overline{BC}$ (Opp. sides of a $\square$ are $\cong$.)
4. $\overline{DC} \cong \overline{CD}$ (Refl. Prop.)
5. $\overline{AC} \cong \overline{BD}$ (Diagonals of a rectangle are $\cong$.)
6. $\triangle ADC \cong \triangle BCD$ (SSS)

21. **Given:** $QTVW$ is a rectangle.

$\overline{QR} \cong \overline{ST}$



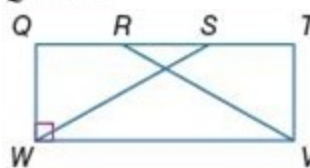
Prove: $\triangle SWQ \cong \triangle RVT$

SOLUTION:

You need to walk through the proof step by step. Look over what you are given and what you need to prove. Here, you are given $QTVW$ is a rectangle and $\overline{QR} \cong \overline{ST}$. You need to prove $\triangle SWQ \cong \triangle RVT$. Use the properties that you have learned about rectangles to walk through the proof.

Given: $QTVW$ is a rectangle.

$\overline{QR} \cong \overline{ST}$



Prove: $\triangle SWQ \cong \triangle RVT$

Proof:

Statements (Reasons)

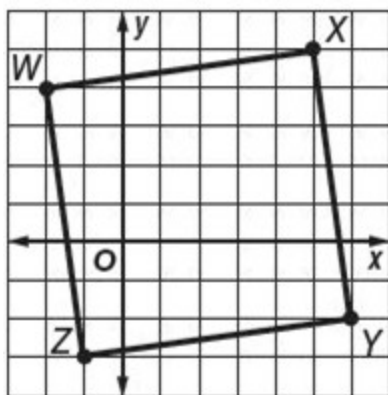
1. $QTVW$ is a rectangle; $\overline{QR} \cong \overline{ST}$. (Given)
2. $QTVW$ is a parallelogram. (Def. of rectangle)
3. $\overline{WQ} \cong \overline{VT}$ (Opp sides of a $\square$ are $\cong$.)
4. $\angle Q$ and $\angle T$ are right angles. (Def. of rectangle)
5. $\angle Q \cong \angle T$ (All rt $\angle$ s are $\cong$.)
6. $QR = ST$ (Def. of $\cong$ segs.)
7. $\overline{RS} \cong \overline{RS}$ (Refl. Prop.)
8. $RS = RS$ (Def. of $\cong$ segs.)
9. $QR + RS = RS + ST$ (Add. prop.)
10. $QS = QR + RS$, $RT = RS + ST$ (Seg. Add. Post.)
11. $QS = RT$ (Subst.)
12. $\overline{QS} \cong \overline{RT}$ (Def. of $\cong$ segs.)
13. $\triangle SWQ \cong \triangle RVT$ (SAS)

8-4 Rectangles

COORDINATE GEOMETRY Graph each quadrilateral with the given vertices. Determine whether the figure is a rectangle. Justify your answer using the indicated formula.

22. $W(-2, 4)$, $X(5, 5)$, $Y(6, -2)$, $Z(-1, -3)$; Slope Formula

SOLUTION:



Use the slope formula to find the slope of the sides of the quadrilateral.

$$m_{WX} = \frac{5-4}{5-(-2)} = \frac{1}{7}$$

$$m_{XY} = \frac{-2-5}{6-5} = -7$$

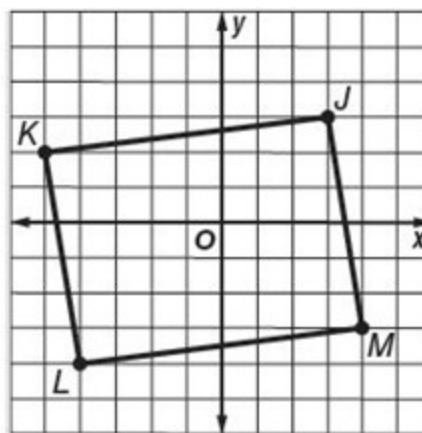
$$m_{YZ} = \frac{-3-(-2)}{-1-6} = \frac{1}{7}$$

$$m_{WZ} = \frac{-3-4}{-1-(-2)} = -7$$

The slopes of each pair of opposite sides are equal. So, the two pairs of opposite sides are parallel. Therefore, the quadrilateral WXYZ is a parallelogram. The products of the slopes of the adjacent sides are -1 . So, any two adjacent sides are perpendicular to each other. That is, all the four angles are right angles. Therefore, WXYZ is a rectangle.

23. $J(3, 3)$, $K(-5, 2)$, $L(-4, -4)$, $M(4, -3)$; Distance Formula

SOLUTION:



First, Use the Distance formula to find the lengths of the sides of the quadrilateral.

$$JK = \sqrt{(-5-3)^2 + (2-3)^2} = \sqrt{(-8)^2 + (-1)^2} = \sqrt{64+1} = \sqrt{65}$$

$$KL = \sqrt{(-4-(-5))^2 + (-4-2)^2} = \sqrt{1^2 + (-6)^2} = \sqrt{37}$$

$$LM = \sqrt{(4-(-4))^2 + (-3-(-4))^2} = \sqrt{8^2 + 1^2} = \sqrt{65}$$

$$JM = \sqrt{(4-3)^2 + (-3-(3))^2} = \sqrt{1^2 + 6^2} = \sqrt{37}$$

$JK = \sqrt{65} = LM$, $KL = 8 = JM$, so JKLM is a parallelogram.

A parallelogram is a rectangle if the diagonals are congruent. Use the Distance formula to find the lengths of the diagonals.

$$KM = \sqrt{(4-(-5))^2 + (-3-2)^2} = \sqrt{9^2 + (-5)^2} = \sqrt{81+25} = \sqrt{106}$$

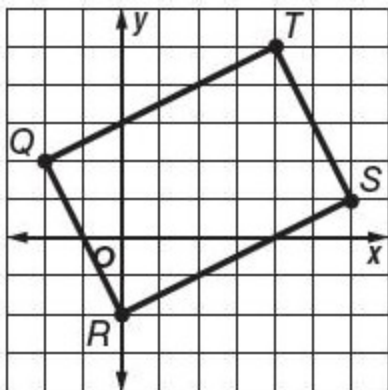
$$JL = \sqrt{(-4-3)^2 + (-4-3)^2} = \sqrt{(-7)^2 + (-7)^2} = \sqrt{49+49} = \sqrt{98}$$

So the diagonals are not congruent. Thus, JKLM is not a rectangle.

8-4 Rectangles

24. $Q(-2, 2)$, $R(0, -2)$, $S(6, 1)$, $T(4, 5)$; Distance Formula

SOLUTION:



First, use the Distance Formula to find the lengths of the sides of the quadrilateral.

$$QR = \sqrt{(0 - (-2))^2 + (-2 - 2)^2} = \sqrt{2^2 + (-4)^2} = \sqrt{4 + 16} = \sqrt{20}$$

$$RS = \sqrt{(6 - 0)^2 + (1 - (-2))^2} = \sqrt{6^2 + 3^2} = \sqrt{36 + 9} = \sqrt{45}$$

$$ST = \sqrt{(4 - 6)^2 + (5 - 1)^2} = \sqrt{(-2)^2 + 4^2} = \sqrt{4 + 16} = \sqrt{20}$$

$$QT = \sqrt{(4 - (-2))^2 + (5 - 2)^2} = \sqrt{6^2 + 3^2} = \sqrt{36 + 9} = \sqrt{45}$$

$QR = \sqrt{20} = ST$, $RS = \sqrt{45} = QT$, so $QRST$ is a parallelogram.

A parallelogram is a rectangle if the diagonals are congruent. Use the Distance Formula to find the lengths of the diagonals.

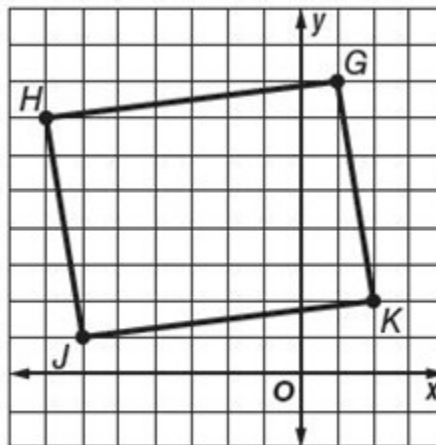
$$QS = \sqrt{(6 - (-2))^2 + (1 - 2)^2} = \sqrt{8^2 + 1^2} = \sqrt{64 + 1} = \sqrt{65}$$

$$RT = \sqrt{(4 - 0)^2 + (5 - (-2))^2} = \sqrt{4^2 + 7^2} = \sqrt{16 + 49} = \sqrt{65}$$

So the diagonals are congruent. Thus, $QRST$ is a rectangle.

25. $G(1, 8)$, $H(-7, 7)$, $J(-6, 1)$, $K(2, 2)$; Slope Formula

SOLUTION:



First, use the slope formula to find the lengths of the sides of the quadrilateral.

$$m_{GH} = \frac{7 - 8}{-7 - 1} = \frac{1}{8}$$

$$m_{HJ} = \frac{1 - 7}{-6 - (-7)} = -6$$

$$m_{JK} = \frac{2 - 1}{2 - (-6)} = \frac{1}{8}$$

$$m_{GK} = \frac{2 - 8}{2 - 1} = -6$$

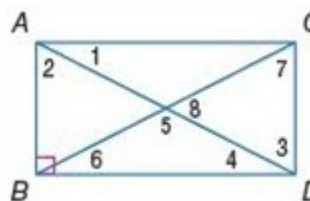
The slopes of each pair of opposite sides are equal.

So, the two pairs of opposite sides are parallel.

Therefore, the quadrilateral $GHJK$ is a parallelogram.

None of the adjacent sides have slopes whose product is -1 . So, the angles are not right angles. Therefore, $GHJK$ is not a rectangle.

Quadrilateral $ABCD$ is a rectangle. Find each measure if $m\angle 2 = 40$.



26. $m\angle 1$

SOLUTION:

All four angles of a rectangle are right angles. So,

$$m\angle 1 + m\angle 2 = 90,$$

$$m\angle 1 = 90 - 40$$

$$= 50$$

8-4 Rectangles

27. $m\angle 7$

SOLUTION:

The measures of angles 2 and 3 are equal as they are alternate interior angles.

Since the diagonals of a rectangle are congruent and bisect each other, the triangle with the angles 3, 7 and 8 is an isosceles triangle. So, $m\angle 7 = m\angle 3$.

Therefore, $m\angle 7 = m\angle 2 = 40$.

28. $m\angle 3$

SOLUTION:

The measures of angles 2 and 3 are equal as they are alternate interior angles.

Therefore, $m\angle 3 = m\angle 2 = 40$.

29. $m\angle 5$

SOLUTION:

All four angles of a rectangle are right angles. So,

$$m\angle 1 + m\angle 2 = 90.$$

$$m\angle 1 = 90 - 40$$

$$= 50$$

The measures of angles 1 and 4 are equal as they are alternate interior angles.

Therefore, $m\angle 4 = m\angle 1 = 50$.

Since the diagonals of a rectangle are congruent and bisect each other, the triangle with the angles 4, 5 and 6 is an isosceles triangle. So, $m\angle 6 = m\angle 4 = 50$.

The sum of the three angles of a triangle is 180.

Therefore, $m\angle 5 = 180 - (50 + 50) = 80$.

30. $m\angle 6$

SOLUTION:

All four angles of a rectangle are right angles. So,

$$m\angle 1 + m\angle 2 = 90.$$

$$m\angle 1 = 90 - 40$$

$$= 50$$

The measures of angles 1 and 4 are equal as they are alternate interior angles.

Therefore, $m\angle 4 = m\angle 1 = 50$.

Since the diagonals of a rectangle are congruent and bisect each other, the triangle with the angles 4, 5 and 6 is an isosceles triangle. Therefore, $m\angle 6 = m\angle 4 = 50$.

31. $m\angle 8$

SOLUTION:

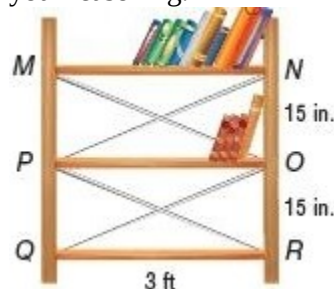
The measures of angles 2 and 3 are equal as they are alternate interior angles.

Since the diagonals of a rectangle are congruent and bisect each other, the triangle with the angles 3, 7 and 8 is an isosceles triangle. So, $m\angle 3 = m\angle 7 = 40$.

The sum of the three angles of a triangle is 180.

Therefore, $m\angle 8 = 180 - (40 + 40) = 100$.

32. **CCSS MODELING** Jody is building a new bookshelf using wood and metal supports like the one shown. To what length should she cut the metal supports in order for the bookshelf to be *square*, which means that the angles formed by the shelves and the vertical supports are all right angles? Explain your reasoning.



SOLUTION:

In order for the bookshelf to be square, the lengths of all of the metal supports should be equal. Since we know the length of the bookshelves and the distance between them, use the Pythagorean Theorem to find the lengths of the metal supports.

Let x be the length of each support. Express 3 feet as 36 inches.

$$15^2 + 36^2 = x^2$$

$$1521 = x^2$$

$$\sqrt{1521} = x$$

$$39 = x$$

Therefore, the metal supports should each be 39 inches long or 3 feet 3 inches long.

8-4 Rectangles

PROOF Write a two-column proof.

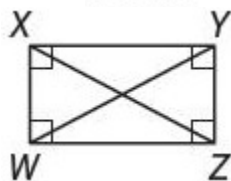
33. Theorem 8.13

SOLUTION:

You need to walk through the proof step by step. Look over what you are given and what you need to prove. Here, you are given $WXYZ$ is a rectangle with diagonals $\overline{WY}$ and $\overline{XZ}$. You need to prove $\overline{WY} \cong \overline{XZ}$. Use the properties that you have learned about rectangles to walk through the proof.

Given: $WXYZ$ is a rectangle with diagonals $\overline{WY}$ and $\overline{XZ}$.

Prove: $\overline{WY} \cong \overline{XZ}$



Proof:

1. $WXYZ$ is a rectangle with diagonals $\overline{WY}$ and $\overline{XZ}$. (Given)
2. $\overline{WX} \cong \overline{ZY}$ (Opp. sides of a $\square$ are $\cong$.)
3. $\overline{WZ} \cong \overline{YZ}$ (Refl. Prop.)
4. $\angle XWZ$ and $\angle YZW$ are right angles. (Def. of rectangle)
5. $\angle XWZ \cong \angle YZW$ (All right $\angle$ s are $\cong$.)
6. $\triangle XWZ \cong \triangle YZW$ (SAS)
7. $\overline{WY} \cong \overline{XZ}$ (CPCTC)

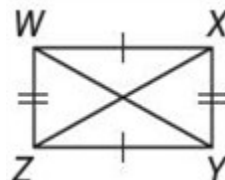
34. Theorem 8.14

SOLUTION:

You need to walk through the proof step by step. Look over what you are given and what you need to prove. Here, you are given $\overline{WX} \cong \overline{YZ}$, $\overline{XY} \cong \overline{WZ}$, and $\overline{WY} \cong \overline{XZ}$. You need to prove that $WXYZ$ is a rectangle. Use the properties that you have learned about rectangles to walk through the proof.

Given: $\overline{WX} \cong \overline{YZ}$, $\overline{XY} \cong \overline{WZ}$, and $\overline{WY} \cong \overline{XZ}$

Prove: $WXYZ$ is a rectangle.



Proof:

1. $\overline{WX} \cong \overline{YZ}$, $\overline{XY} \cong \overline{WZ}$, and $\overline{WY} \cong \overline{XZ}$ (Given)
2. $\triangle WZX \cong \triangle XYW$ (SSS)
3. $\angle ZWX \cong \angle YXW$ (CPCTC)
4. $m\angle ZWX = m\angle YXW$ (Def. of $\cong \angle$ s)
5. $WXYZ$ is a parallelogram. (If both pairs of opp. sides are $\cong$, then quad. is $\square$.)
6. $\angle ZWX$ and $\angle YXW$ are supplementary. (Cons. $\angle$ s of $\square$ are suppl.)
7. $m\angle ZWX + m\angle YXW = 180$ (Def of suppl.)
8. $\angle ZWX$ and $\angle YXW$ are right angles. (If 2 $\angle$ are $\cong$ and suppl., each $\angle$ is a rt. $\angle$.)
9. $\angle WZY$ and $\angle XYZ$ are right angles. (If a $\square$ has 1 rt. $\angle$, it has 4 rt. $\angle$ s.)
10. $WXYZ$ is a rectangle. (Def. of rectangle)

8-4 Rectangles

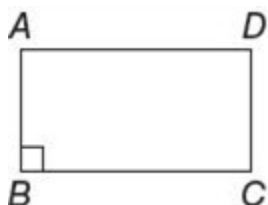
PROOF Write a paragraph proof of each statement.

35. If a parallelogram has one right angle, then it is a rectangle.

SOLUTION:

You need to walk through the proof step by step.

Look over what you are given and what you need to prove. Here, you are given a parallelogram with one right angle. You need to prove that it is a rectangle. Let $ABCD$ be a parallelogram with one right angle. Then use the properties of parallelograms to walk through the proof.



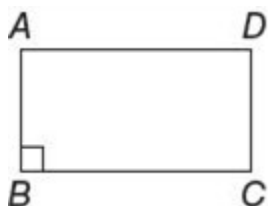
$ABCD$ is a parallelogram, and $\angle B$ is a right angle. Since $ABCD$ is a parallelogram and has one right angle, then it has four right angles. So by the definition of a rectangle, $ABCD$ is a rectangle.

36. If a quadrilateral has four right angles, then it is a rectangle.

SOLUTION:

You need to walk through the proof step by step.

Look over what you are given and what you need to prove. Here, you are given a quadrilateral with four right angles. You need to prove that it is a rectangle. Let $ABCD$ be a quadrilateral with four right angles. Then use the properties of parallelograms and rectangles to walk through the proof.



$ABCD$ is a quadrilateral with four right angles. $ABCD$ is a parallelogram because both pairs of opposite angles are congruent. By definition of a rectangle, $ABCD$ is a rectangle.

37. **CONSTRUCTION** Construct a rectangle using the construction for congruent segments and the construction for a line perpendicular to another line through a point on the line. Justify each step of the

construction.

SOLUTION:

Sample answer:

Step 1: Graph a line and place points P and Q on the line.

Step 2: Place the compass on point P . Using the same compass setting, draw an arc to the left and right of P that intersects _____.

Step 3: Open the compass to a setting greater than the distance from P to either point of intersection. Place the compass on each point of intersection and draw arcs that intersect above $\overline{PQ}$.

Step 4: Draw line a through point P and the intersection of the arcs drawn in step 3.

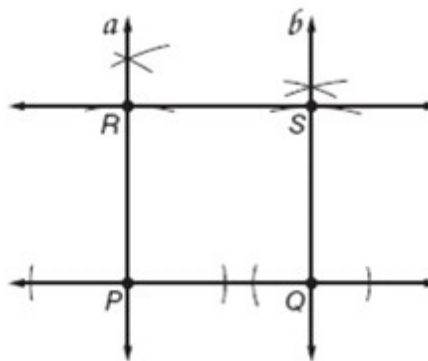
Step 5: Repeat steps 2 and 3 using point Q .

Step 6: Draw line b through point Q and the intersection of the arcs drawn in step 5.

Step 7: Using any compass setting, put the compass at P and draw an arc above it that intersects line a . Label the point of intersection R .

Step 8: Using the same compass setting, put the compass at Q and draw an arc above it that intersects line b . Label the point of intersection S .

Step 9: Draw $\overline{RS}$.



Since $\overline{RP} \perp \overline{PQ}$ and $\overline{SQ} \perp \overline{PQ}$, the measure of angles P and Q is 90 degrees. Lines that are perpendicular to the same line are parallel, so $\overline{RP} \parallel \overline{SQ}$. The same compass setting was used to